Commissioned First-Use Evaluation of Metriplane 0.3.0 Across Six Computing Environments

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Evaluated software	metriplane==0.3.0 from PyPI
Protocol	MP-EXT-UX-001 v1.0
Evaluated tag commit	e8ee6c63deaee47bd450c5d6c7523d5bd699852a
Source data commit	d2a13b800853a2acea1240ec8af6c34ef442dd77
Publication status	Technical report; not peer reviewed

Abstract
This report evaluates the first-use experience of Metriplane 0.3.0 using a predefined, camera-free packaged workflow. Six compensated external evaluators executed the workflow on independently controlled computing environments spanning Native Linux, macOS, WSL2 Ubuntu, x86_64, ARM64/AArch64, Python 3.12, and Python 3.13. Compensation was fixed independently of the evaluation outcome. All six records accepted under the frozen protocol completed the bounded workflow. Two additional commissioned executions were retained separately because their first attempts did not satisfy the protocol requirements. The evaluation also identified WSL2-specific report-access and documentation friction, which resulted in a subsequent documentation change. The results characterize this specific first-use workflow and the observed environments. They do not constitute a general reliability, compatibility, safety, physical-accuracy, adoption, or production-readiness evaluation.

6/6	6	2	1
accepted records completed the bounded workflow	observed computing environments	auxiliary executions reported separately	WSL2 documentation cluster identified

1. Background and objective

Metriplane is an observe-only research-software toolkit for turning recorded workcell state into inspectable incidents, integrity-verifiable evidence bundles, and generated regression checks. The purpose of this evaluation was narrower than a product benchmark or deployment study. It asked whether a first-time external evaluator, using an eligible pre-existing environment, could install Metriplane 0.3.0 from PyPI and complete the packaged camera-free workflow without maintainer troubleshooting before the first-attempt evidence was preserved.

The study was designed to produce an auditable record of first-use behavior across a deliberately varied set of computing environments. It did not evaluate camera calibration, physical tracking accuracy, a live robot, a factory process, safety behavior, or compatibility with arbitrary robotics data.

2. Evaluated software and workflow

The evaluated artifact was `metriplane==0.3.0`, installed from PyPI. The v0.3.0 release tag resolves to commit `e8ee6c63deaee47bd450c5d6c7523d5bd699852a`.

```
metriplane --version
metriplane doctor
metriplane demo
```

The packaged demo required no camera, GPU, robot, ROS installation, Docker installation, or external dataset. It generated an incident timeline and report, an evidence bundle that could be verified, and a generated regression check that could be executed. Opening the HTML report in a browser was optional and was not part of the core completion criterion.

3. Study design

3.1 Recruitment and compensation

6. Auxiliary commissioned executions

Two additional executions produced useful technical information but did not satisfy the frozen first-attempt rules.

Record	Initial status	Later result	Reason for exclusion
AUX-A	Invalid precondition	Follow-up completed	WSL2 and the required Python environment were provisioned during execution, and work continued after failed preconditions.
AUX-B	Failed first attempt	Second run completed	Virtual-environment creation failed in a non-writable Windows system directory. Written working-directory guidance was supplied and a second run was performed.

The later successful technical results are retained in `auxiliary_records.csv`. They are not included in the six-record primary denominator.

7. Usability finding and engineering response

The WSL2 records identified a small documentation and report-access cluster. A Windows user could complete the core workflow but still pause when translating a Linux output path into a Windows-accessible path. The stable filename `cell_truth_report.html` also did not immediately communicate that it was the generated incident report.

The observation was documented publicly in GitHub issue #60. PR #61 added WSL2 writable-directory guidance, explicit output-path guidance, Windows report-opening instructions, and stable-filename guidance. The documentation-only change was merged as commit `5e5396f0756b47ff02f6f0831ec62a44ea21c118`.

WSL2 observations	GitHub issue #60	PR #61	Documentation commit 5e5396f
Report access and working-directory friction	Public finding and scope	Documentation-only response	Merged without runtime change

The response did not change Metriplane runtime behavior, the report filename, evidence verification, generated regression behavior, the v0.3.0 tag, or the original cohort. No post-fix execution was added to the v0.3.0 result denominator.

8. Interpretation and limitations

This evaluation provides evidence about a specific packaged first-use path under recorded conditions. It shows that six compensated external evaluators completed that path in six observed environments accepted under the frozen protocol. It also shows that protocol-invalid executions can reveal useful usability information without being counted as primary passes.

- The sample was small, selected, compensated, and not random.
- The result does not measure organic adoption, customer demand, or institutional use.
- Only the packaged camera-free workflow was evaluated.
- No live camera, physical measurement, robot, factory, control system, or safety function was evaluated.
- No generic ROS 2, Isaac Sim, robotics-data, Linux, macOS, ARM64, or WSL2 compatibility claim follows from these observations.
- Some structured fields were not recorded for all records, most notably T02.
- Browser dispatch from WSL2 remains dependent on the local Windows and WSL configuration.
- Successful first use does not establish production readiness or absence of other defects.

9. Data availability and reproducibility

The publication package contains the canonical report source, rendered PDF, de-identified primary and auxiliary tables, a machine-readable aggregate summary, a public source manifest, and integrity hashes.

File	Purpose
REPORT.md	Canonical human-readable report
commissioned_first_use_evaluation.pdf	Rendered publication
public_results.csv	Six accepted primary records
auxiliary_records.csv	Two excluded auxiliary records
summary.json	Machine-readable aggregate values
SOURCE_MANIFEST.json	Public source provenance
SHA256SUMS	Integrity hashes for the package

The public tables use cohort labels rather than evaluator names. Public GitHub result links are retained for technical traceability. Marketplace orders, billing records, legal identities, private messages, and private administrative records are not distributed because they contain personal or financial information.

The report package was prepared from Metriplane repository merge commit `d2a13b800853a2acea1240ec8af6c34ef442dd77`.

10. Conclusion

Six compensated external evaluators produced six accepted primary records for the same frozen Metriplane 0.3.0 packaged first-use workflow. All six accepted records completed the bounded workflow in their observed environments. Two additional commissioned executions were retained separately because their initial execution histories did not satisfy the protocol. The study also identified WSL2-specific report-access and documentation friction that led to a documentation-only engineering response.

The result should be interpreted as a bounded first-use evaluation, not as evidence of general reliability, adoption, physical accuracy, safety, broad compatibility, or production readiness.

Public references

1. **Metriplane v0.3.0 release.** https://github.com/Miko997/metriplane/releases/tag/v0.3.0
2. **Public commissioned-result thread.** https://github.com/Miko997/metriplane/issues/27
3. **WSL2 documentation finding.** https://github.com/Miko997/metriplane/issues/60
4. **Documentation response.** https://github.com/Miko997/metriplane/pull/61
5. **Documentation merge commit.** https://github.com/Miko997/metriplane/commit/5e5396f0756b47ff02f6f0831ec62a44ea21c118

Interpretation note. This is a technical report of a commissioned first-use evaluation. It has not undergone peer review and should be cited and interpreted within the stated study boundary.